

Sardar Vallabhbhai National Institute of Technology, Surat
Computer Science and Engineering Department
M.Tech(CSE/DS)-I - Sem-I - 2026-27
Computer Vision and Image Processing(CSCS111/CSDS119)
Lab Assignment 1: Understanding and Analysing Digital Images Using C

Problem Statement:

Take two different grayscale images of your choice and write a C program to read and analyse their pixel values. Use the obtained information to compare the two images and draw suitable conclusions.

- Read both images and store their pixel values in arrays.
- Determine the dimensions, minimum intensity, maximum intensity, and average intensity of each image.
- Generate the histogram of both images and analyse the distribution of their pixel intensities.
- Divide the intensity values into suitable ranges and determine the percentage of dark, medium, and bright pixels.
- Identify which image is brighter, darker, or has higher contrast based on numerical analysis rather than only visual observation.
- Select a region from each image having significantly different intensity from the rest of the image and analyse that region separately.
- Modify the pixel values of one image to make it brighter without changing its dimensions. Analyse the changes in its pixel statistics and histogram.
- Investigate whether two visually different images can have the same average intensity. Demonstrate your observation using suitable images or pixel values.
- Based on the obtained statistics and histograms, determine which image would be more suitable for further processing such as thresholding and justify your answer.

Requirement: The analysis should be performed using arrays and basic C operations. Display the computed statistics and histograms and provide brief reasoning for the conclusions drawn from the pixel-level analysis.